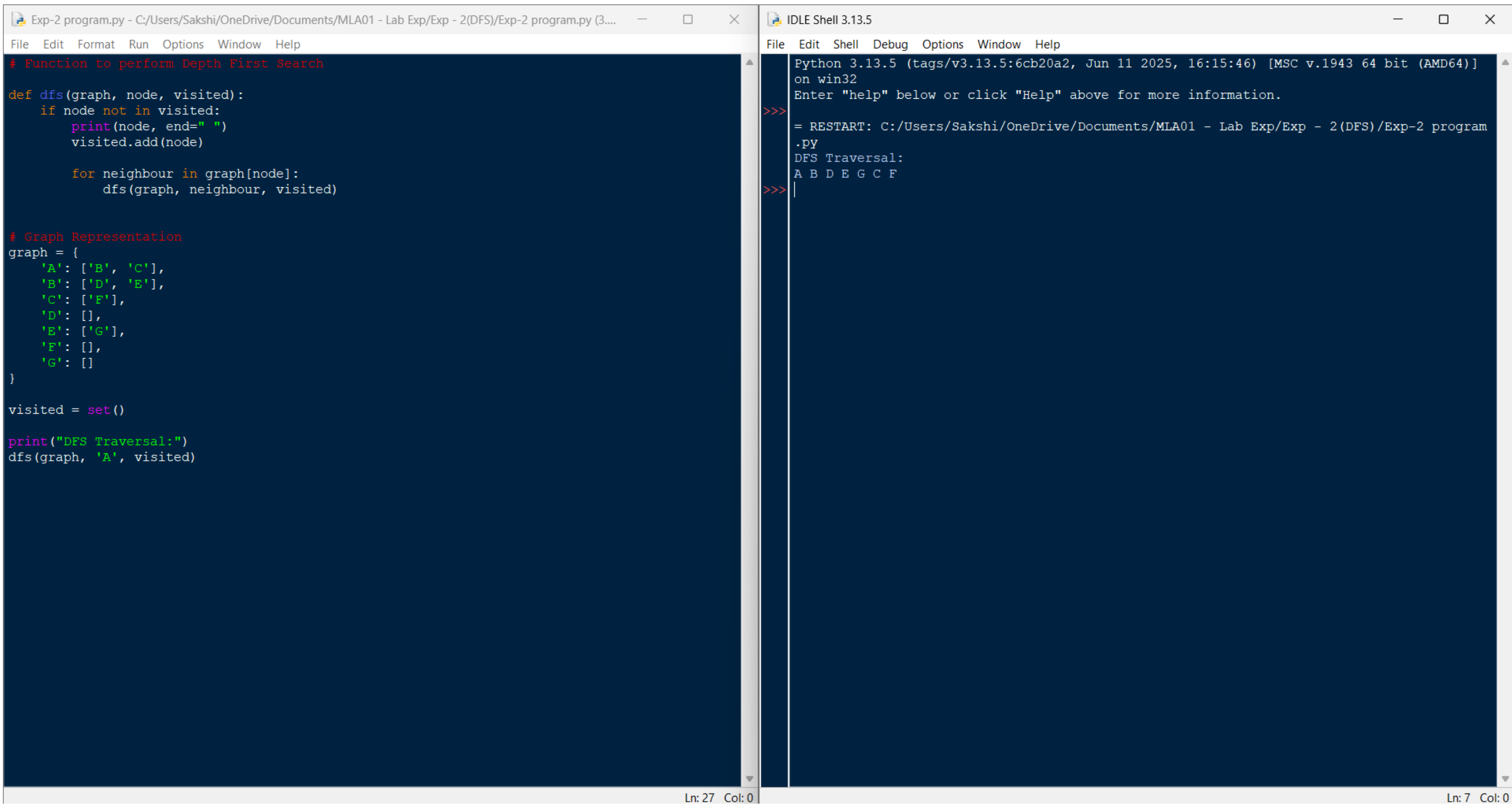
Exp:2 DFS

Program:

```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for neighbour in graph[node]:
            dfs(graph, neighbour, visited)
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}
visited = set()
print("DFS Traversal:")
dfs(graph, 'A', visited)
```

Output:


Exp-2 program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp - 2(DFS)/Exp-2 program.py (3....
File Edit Format Run Options Window Help
# Function to perform Depth First Search
def dfs(graph, node, visited):
if node not in visited:
print(node, end=" ")
visited.add(node)
for neighbour in graph[node]:
dfs(graph, neighbour, visited)
# Graph Representation
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F'],
'D': [],
'E': ['G'],
'F': [],
'G': []
}
visited = set()
print("DFS Traversal:")
dfs(graph, 'A', visited)
Ln: 27 Col: 0
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp - 2(DFS)/Exp-2 program.py
DFS Traversal:
A B D E G C F
Ln: 7 Col: 0